

REVIEW

The enigmatic function of IgD: some answers at last

Cindy Gutzeit¹, Kang Chen^{2,3} and Andrea Cerutti^{1,3,4,5} ¹ Immunology Institute, Department of Medicine, Mount Sinai Icahn School of Medicine, New York, NY, USA² Department of Obstetrics and Gynecology, Wayne State University School of Medicine, Detroit, MI, USA³ Mucosal Immunology Studies Team (MIST), National Institute of Allergy and Infectious Diseases, National Institutes of Health, Bethesda, MD, USA⁴ Program for Inflammatory and Cardiovascular Disorders, Institut Hospital del Mar d'Investigacions Mèdiques (IMIM), Barcelona, Spain⁵ Catalan Institute for Research and Advanced Studies (ICREA), Barcelona, Spain

IgD emerged soon after IgM at the time of inception of the adaptive immune system. Despite its evolutionary conservation from fish to humans, the specific functions of IgD have only recently begun to be elucidated. Mature B cells undergo alternative mRNA splicing to express IgD and IgM receptors with identical antigenic specificity. The enigma of dual IgD and IgM expression has been tackled by several recent studies showing that IgD helps peripheral accumulation of physiologically autoreactive B cells through its functional unresponsiveness to self-antigens but prompt readiness against foreign antigens. IgD achieves this balance by attenuating IgM-mediated anergy while promoting specific responses to multimeric non-self-antigens. Additional research has clarified how and why certain mucosal B cells become plasmablasts or plasma cells specializing in IgD secretion. In particular, the microbiota has been shown to play an important role in driving class switch-mediated replacement of IgM with IgD. Secreted IgD appears to enhance mucosal homeostasis and immune surveillance by “arming” myeloid effector cells such as basophils and mast cells with IgD antibodies reactive against mucosal antigens, including commensal and pathogenic microbes. Here we will review these advances and discuss their implications in humoral immunity in human and mice.

Keywords: Antibodies · B cells · BCR · Immune responses · Mucosal immunity

Introduction

B cells from both humans and mice mediate humoral immunity by producing IgM, IgD, IgG, IgA and IgE [1]. These antibodies can be either expressed as cell surface signal-transducing antigen receptors (BCRs) or secreted as soluble molecules [2]. While the biology of IgM, IgG, IgA and IgE has been extensively studied, the regulation and function of IgD have, until recently, remained

less well understood [3, 4]. Initially considered a recently evolved antibody, IgD was later identified in bony and cartilaginous fishes, which appeared about 500 million years ago, when the adaptive immune system first evolved [5]. The preservation of IgD throughout evolution implies that IgD exerts important functions that confer a specific survival advantage to the host [6].

In both humans and rodents, the majority of antigen-naïve mature B cells dually express surface IgD and IgM through a mechanism involving alternative splicing of a heterogeneous precursor mRNA (pre-RNA) in the nucleus [7, 8]. However, a small subset of mucosal B cells from nasopharyngeal lymphoid tissues expresses IgD but not IgM. These cells derive from an unconventional

Correspondence: Dr. Andrea Cerutti
e-mail: acerutti@imim.es

form of class switch DNA recombination (CSR) and thereafter differentiate into plasmablasts or plasma cells exclusively secreting IgD [9–14]. In humans, secreted IgD can be detected in the circulation, nasopharyngeal, oral and lachrymal secretions, as well as on the surface of myeloid innate effector cells, including basophils, mast cells and monocytes [12, 15, 16]. The scope of this review is to highlight recent advances in our understanding of the regulation and function of transmembrane and secreted IgD in both humans and mice.

Regulation of transmembrane IgD expression

BCR expression is essential for the development, survival and activation of murine as well as human B cells [17]. Stochastic recombination-activating gene (RAG)-mediated assembly of Ig variable (V), diversity (D) and/or joining (J) gene segments in pro-B and pre-B cells from the bone marrow leads to the development of immature B cells expressing fully functional IgM receptors made up of complete heavy chain (HC) and light chain (LC) heterotetramers [18]. Besides generating antigen recognition diversity, this early antigen-independent phase of B-cell ontogeny encompasses tolerance checkpoints that remove self-reactive BCRs by clonal deletion and receptor editing, so that only immature B cells that do not recognize autoantigens leave the bone marrow as transitional B cells [19, 20]. **However, earlier mouse studies showed that a significant fraction of self-reactive B cells exited the bone marrow to join the normal peripheral B cell repertoire as anergic B cells** [21, 22]. These initial studies have been recently confirmed by research performed in both humans and mice which elucidated the mechanisms underpinning the formation and function of self-reactive anergic B cells inhabiting peripheral lymphoid organs [23–25]. Such mechanisms may involve the activation of an anergy-inducing B cell signaling program upon BCR engagement by endogenous self-antigen [24–26]. As discussed later, these mechanisms probably evolved to preserve a peripheral BCR repertoire, which exhibits broad reactivity and is thereby capable of eliminating microbes that mimic self-antigens [26–28]. In the spleen, IgM-expressing transitional B cells further differentiate into mature follicular (FO) and marginal zone (MZ) B cells through distinct maturation pathways characterized by progressive up-regulation of IgD receptors [29, 30]. Dual expression of surface IgM and IgD with identical specificities occurs through differential polyadenylation and alternative splicing of a 25-kb-long heterogeneous pre-RNA encompassing the recombined V_HDJ_H exon and the downstream C_μ and C_δ exons encoding the heavy chain constant (C) regions of IgM and IgD (Fig. 1), respectively [7, 31].

Recent mouse studies have shown that IgD expression by transitional and mature B cells requires zinc-finger protein 318 (ZFP318), a nuclear protein with two U1-type zinc fingers usually associated with RNA-binding factors [8, 32]. ZFP318 expression has been found to follow a pattern similar to that of IgD, as pro-B and pre-B cells express little or no ZFP318, whereas imma-

ture, transitional and mature B cells progressively up-regulate ZFP318 [8, 32]. A deleterious isoleucine-to-threonine substitution within a conserved lysine-rich IgD-promoting domain in ZFP318 abolishes surface IgD expression on MZ B cells and decreases surface IgD expression on FO B cells without perturbing the accumulation of either B cell subset in the spleen [8]. Although MZ and FO B cells from ZFP318 mutant mice exhibit increased IgM expression, they show normal surface patterns of CD21, CD23 and CD62L, a set of molecules which is transcriptionally regulated in parallel with IgD during B cell development [8]. Both ZFP318-deficient and ZFP318 mutant mice show conserved splenic pools of MZ and FO B cells that completely lack surface IgD. However, neither of these mice display additional major phenotypic alterations besides increased IgM expression [8].

ZFP318 does not affect B cell development, but instead promotes B cell expression of IgD by directly regulating the alternative splicing of C_μ and C_δ exons from nuclear pre-mRNA (Fig. 1). This process may involve pre-mRNA interaction with ZFP318, as this protein preferentially accumulates in the nucleus and carries zinc finger and lysine-rich domains similar to those in U1 small nuclear ribonuclear protein C, a nuclear RNA-binding protein that assembles the spliceosome machinery at the 5' splice-site of pre-mRNAs [8]. Further studies from mice lacking ZFP318 in hematopoietic cells have suggested that ZFP318 also interferes with the recognition of a transcriptional stop site located at the 3' end of the terminal C_μ -encoding exon [32]. This blocking effect would allow for the RNA polymerase II complex to advance through the IgD-encoding region of the IgH locus, thereby permitting the synthesis of a pre-RNA encompassing both C_μ and C_δ exons (Fig. 1). This pre-mRNA is required for IgD and IgM co-expression [32]. Although more studies will be needed to further clarify how ZFP318 regulates C_μ and C_δ gene transcription and alternative C_μ and C_δ mRNA splicing, these findings provide long-sought evidence of a *trans*-acting factor essential for IgD expression.

Regulation of IgD class switching

In humans, some IgM⁺IgD⁺ FO B cells from tonsils lose IgM but retain IgD expression after undergoing CSR through a mechanism involving activation-induced cytidine deaminase (AID) [9–12]. The resulting IgM[−]IgD⁺ B cells differentiate into IgD-secreting plasma cells [12], some of which enter the general circulation to further disseminate into nasal cavities, lachrymal glands, and salivary glands but not intestine [12, 33, 34]. This specific topography may result from the expression of a discrete set of tissue-homing receptors by IgM[−]IgD⁺ B cells and their plasma cell progeny [35]. Interestingly, IgM[−]IgD⁺ B cells have been also identified in channel catfish, where they secrete IgD consisting of a δ HC pairing with an uncommon σ LC only rarely associated with other HCs [36]. Similarly, human but not mouse IgM[−]IgD⁺ B cells preferentially utilize λ LC instead of the more common κ LC [9, 12], which may hint to a derivation of human IgM[−]IgD⁺ B cells from a unique λ -biased B-cell lineage committed to undergo IgM-to-IgD CSR. The

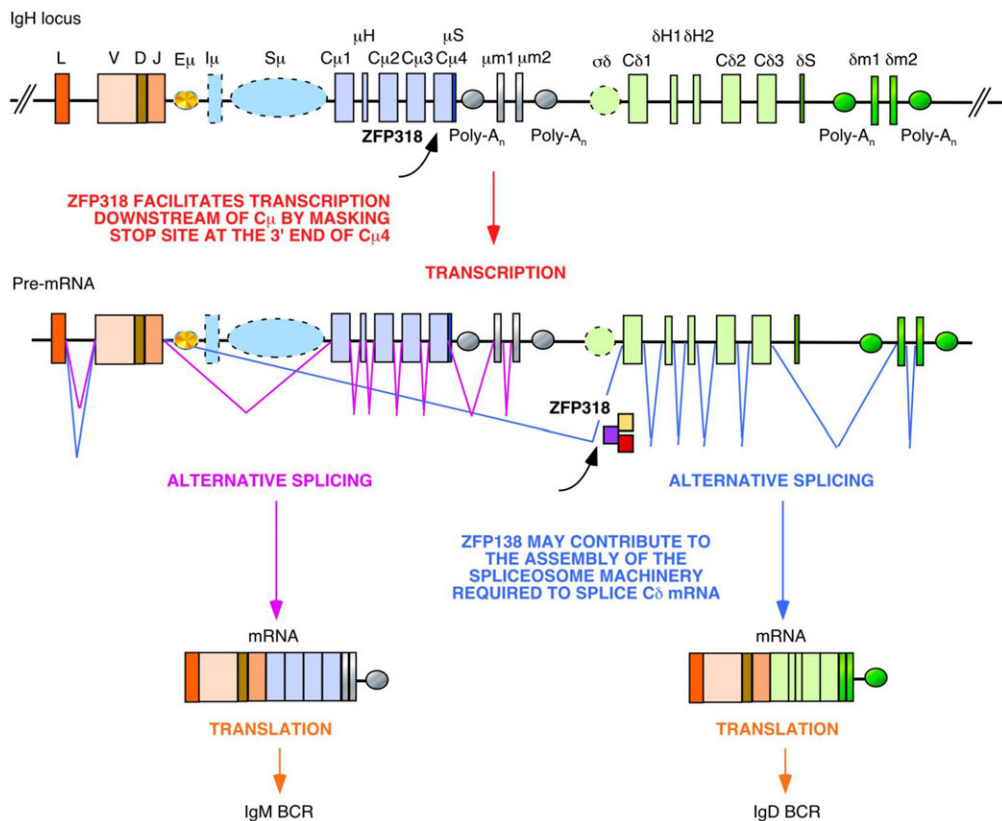


Figure 1. Schematics of alternative splicing for surface IgM and IgD co-expression. Mature B cells constitutively transcribe VDJ, C μ and C δ genes to form a long pre-mRNA through a process involving ZFP318. This protein facilitates C δ gene transcription by masking a stop site in the terminal portion of the C μ gene. The resulting pre-mRNA undergoes alternative splicing to remove intronic regions and promote dual expression of mature VDJ-C μ and VDJ-C δ mRNAs encoding IgM and IgD receptors. ZFP318 may also facilitate the splicing of C δ exons included in the pre-mRNA, probably by promoting the assembly of the spliceosome machinery at the 5' start site of C δ exons. Graphics depict a segment of the human IgH locus and the resulting pre-mRNA and mature mRNAs. L, leader sequence. E μ , IgH enhancer μ , also known as IgH 5' enhancer or IgH intronic enhancer. I μ , initiation μ exon. μ S, 3' segment of the C μ 4 exon encoding the secretory (S) piece of IgM. Poly-A $_n$, terminal polyadenylation exon. μ m, exon encoding the transmembrane region of IgM. $\sigma\delta$, cryptic switch region. δ H, exon encoding the hinge (H) region of IgD. δ S, exon encoding the secretory piece of IgD. δ m, exon encoding the transmembrane region of IgD.

mechanisms underlying this process have been partly elucidated by recent studies [13, 14].

Classical CSR replaces the C μ gene encoding the C H region of IgM with C γ , C α or C ϵ genes encoding the C H region of IgG, IgA or IgE, respectively [37] (Fig. 2). This process enables B cells to replace IgM with downstream antibody classes characterized by identical specificity but different effector functions [37]. Similar to somatic hypermutation (SHM), CSR requires the DNA-editing enzyme AID, which targets the repetitive, G-rich S regions positioned upstream of each C H region [37]. AID targeting of S regions is preceded by cytokine-mediated induction of germline transcription (Fig. 2), which is controlled by the IgH 3' regulatory region (3'RR), a complex transcriptional enhancer located downstream of human and mouse C α genes [38]. Germline transcription generates a primary transcript that physically associates with the template strand of the S region DNA to form a stable DNA-RNA hybrid [37]. This structure generates R loops, in which the displaced non-template strand exists as a G-rich single-stranded DNA [37]. AID deaminates cytosine residues on both strands of S region DNA, thereby generating multiple DNA lesions that are ultimately pro-

cessed into double-stranded DNA breaks [37]. Fusion of double-stranded DNA breaks at donor and acceptor S regions through the classical non-homologous end-joining pathway induces looping-out deletion of the intervening DNA [37], thereby juxtaposing the recombined V H DJ H gene encoding the antigen-binding V H region of the rearranging Ig molecule to a new C H gene (Fig. 2).

Unlike C γ , C α and C ϵ genes, the C δ gene is not preceded by a canonical S region despite being constitutively transcribed together with the C μ gene. This observation led researchers to initially believe that the generation of IgM⁺ IgD⁺ B cells does not involve CSR [37]. However, a cryptic, short intronic region, $\sigma\delta$, located between C μ and C δ genes in both human and cow IgH loci (Fig. 3) was later shown to function as an acceptor site for the donor S μ region to mediate non-homologous IgM-to-IgD CSR [9, 12, 39]. Additional work has subsequently identified I μ and $\Sigma\mu$, two identical and very short intronic regions found upstream and downstream of C μ in human and mouse IgH loci, respectively. It is likely that these regions are also capable of mediating homologous IgM-to-IgD CSR, analogously to $\sigma\delta$ [40]. In humans,

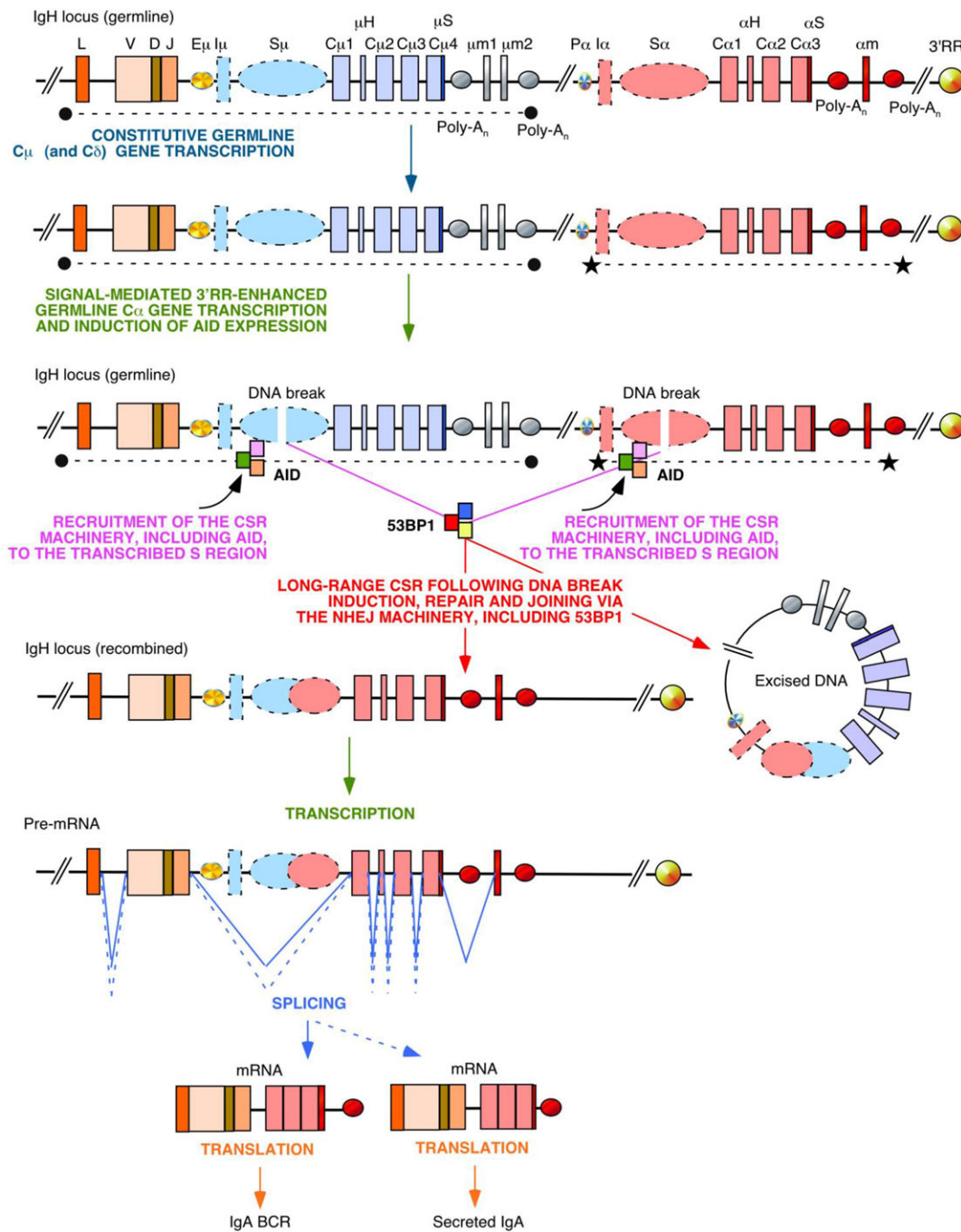


Figure 2. Schematics of classical CSR from IgM to IgA. Mature B cells constitutively transcribe VDJ, C μ and C δ genes encoding IgM and IgD. In the presence of specific sets of activation signals, B cells initiate germline transcription of downstream C γ , C α (shown here) or C ϵ genes throughout a genomic region encompassing the initiation (I) exon and the S region located upstream of each of these C H genes. Germline transcription is preceded by the activation of a promoter sequence positioned upstream of the I exon and further requires the 3'RR located downstream of the C α gene (not shown). In addition to initiating germline transcription, B cell-activation signals induce the expression of the DNA-editing enzyme AID, which is an essential component of the CSR machinery. Together with other proteins, AID binds DNA-RNA hybrids from the targeted donor S μ and acceptor S γ , S α (shown here) or S ϵ regions and thereafter introduces DNA lesions that are ultimately processed into double-strand DNA breaks. These breaks undergo fusion through the non-homologous end-joining (NHEJ) pathway, which includes the DNA repair protein 53BP1. The resulting looping-out deletion of the DNA interposed between S μ and S γ , S α (shown here) or S ϵ regions causes loss of the C μ gene and juxtaposition of the VDJ gene with a downstream C γ , C α (shown here) or C ϵ gene. Finally, the recombined IgH locus undergoes transcription and splicing events that lead to the formation of mature mRNAs encoding transmembrane or secreted IgG, IgA (shown here) or IgE. Graphics depict a segment of the human IgH locus and the resulting pre-mRNA and mature mRNAs. L, leader sequence. E μ , IgH enhancer μ , also known as IgH 5' enhancer or IgH intronic enhancer. I μ , initiation μ exon. μ S, 3' segment of the C μ 4 exon encoding the secretory (S) piece of IgM. Poly-A $_n$, terminal polyadenylation exon. μ m, exon encoding the transmembrane region of IgM. P α , promoter. I α , initiation α exon. α S, 3' segment of the C α 3 exon encoding the secretory (S) piece of IgA. α m, exon encoding the transmembrane region of IgA. 3'RR, IgH 3' regulatory region, also known as IgH 3' enhancer.

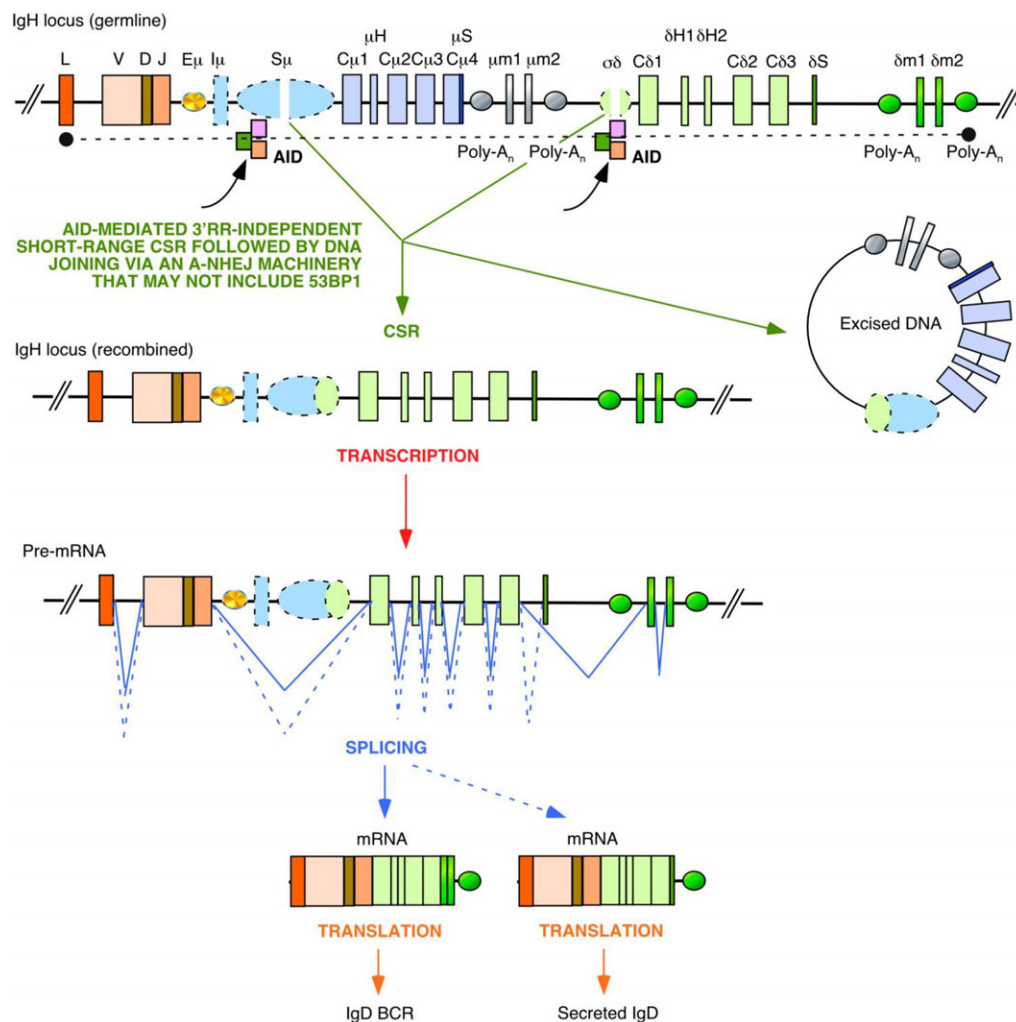


Figure 3. Schematics of non-classical CSR from IgM to IgD. A poorly understood subset of mature B cells recruits the CSR machinery, including AID, to constitutively transcribed S μ and σ δ regions. This process does not require the 3'RR. AID introduces DNA lesions that are ultimately processed into double-strand DNA breaks. These breaks undergo fusion through an alternative non-homologous end-joining (A-NHEJ) pathway that does not require the DNA repair protein 53BP1. The resulting looping-out deletion of the DNA interposed between S μ and σ δ causes loss of the C μ gene and juxtaposition of the VDJ gene with the C δ gene. Finally, the recombined IgH locus undergoes transcription and splicing events that lead to the formation of mature mRNAs encoding transmembrane or secreted IgD. Graphics depict a segment of the human IgH locus and the resulting pre-mRNA and mature mRNAs. L, leader sequence. E μ , IgH enhancer μ , also known as IgH 5' enhancer or IgH intronic enhancer. I μ , initiation μ exon. μ S, 3' segment of the C μ 4 exon encoding the secretory (S) piece of IgM. Poly-A $_n$, terminal polyadenylation exon. μ m, exon encoding the transmembrane region of IgM. σ δ , cryptic switch region. δ H, exon encoding the hinge (H) region of IgD. δ S, exon encoding the secretory piece of IgD. δ m, exon encoding the transmembrane region of IgD.

the involvement of CSR in the generation of IgM⁺IgD⁺ B cells is consistent with the observation that circulating IgM⁺IgD⁺ B cells virtually disappear in primary immunodeficiency patients harboring deleterious AID substitutions [12]. Of note, naïve IgM⁺IgD⁺ B cells from these AID-deficient patients also fail to undergo IgM-to-IgD CSR in response to polyclonal stimulation by exogenous stimuli [12]. The key role played by AID in IgM-to-IgD CSR has been recently established in murine studies [13, 14].

Recently, σ δ has also been identified in the mouse IgH locus, where it recombines with S μ to execute IgM-to-IgD CSR [13, 14]. In mice, this process occurs in a small number of B cells from nasal-associated lymphoid tissue and mesenteric lymph nodes [13, 14]. This latter location is somewhat surprising, because

human IgM⁺IgD⁺ B cells lack gut-homing α 4 β 7 and CCR9 receptors, which are usually induced following processing of dietary vitamin A into retinoic acid by antigen-primed, mesenteric lymph node dendritic cells [35, 41]. The link between IgM-to-IgD CSR and the gut mucosa has been confirmed in mice by a recent study showing targeting of commensal bacteria living in the intestine by secreted IgD [14]. In this study, secreted IgD was shown to decrease in the absence of microbial sensing by immune cells via germline-encoded Toll-like receptors [14]. Similarly, IgD secretion decreases in germ-free or antibiotic-treated mice [14]. These observations suggest that, similarly to IgA, secreted IgD establishes a complex and possibly mutualistic relationship with commensals [41].

Both human and mouse IgM⁺ IgD⁺ B cells recombine S μ with $\sigma\delta$ through an unconventional CSR pathway requiring AID [12–14]. Unlike classical IgM-to-IgG/IgA/IgE CSR, non-classical IgM-to-IgD CSR does not require the 3'RR (Fig. 3) and mostly involves an alternative DNA end-joining pathway rather than the non-homologous DNA end-joining pathway [13, 14]. Accordingly, mice lacking p53-binding protein 1 (53BP1), which mediates non-homologous DNA end joining in the context of classical CSR, show increased IgM-to-IgD CSR and IgD secretion, perhaps to compensate for the lack of IgG, IgA and IgE induction by classical 53BP1-dependent CSR [14]. This finding has led to the proposal that 53BP1 is dispensable for short-range S μ -to- $\sigma\delta$ CSR (Fig. 3). Further work will be needed to clarify how the constitutively transcribed $\sigma\delta$ region recruits AID, what the mechanistic driver of 53BP1-independent IgM-to-IgD CSR is, and whether IgM-to-IgD CSR targets B-1 cells in addition to B-2 cells. In this regard, the enriched autopolyreactivity of human IgM⁺ IgD⁺ B cells is reminiscent of that of murine B-1 cells [10, 11, 42]. However, a remarkable difference between mouse and human IgM⁺ IgD⁺ B cells is that the former express unmutated Ig V(D)J genes, whereas the latter express highly mutated Ig V(D)J genes [9–11, 13]. Given that murine B-1 cells are largely unmutated [42], the derivation of human IgM⁺ IgD⁺ B cells from a B-1 cell-like precursor seems highly unlikely.

Function of transmembrane IgD

IgD receptor and B cell development

Early studies involving δ HC-deficient and μ HC-deficient mice lacking IgD and IgM receptors, respectively, suggested that these BCR isotypic variants are functionally interchangeable [43–46]. In particular, the analysis of BCR transgenic mice expressing hen egg lysozyme (HEL)-reactive IgM alone, IgD alone or both IgD and IgM simultaneously have revealed that IgD and IgM receptors show no major differences in HEL-mediated induction of B-cell tolerance and activation, including CD80 and CD86 up-regulation and antibody secretion [43, 47]. However, other observations argue against a complete functional redundancy of IgD and IgM receptors. For instance, IgD deficiency spares B-cell development but delays affinity maturation during primary antibody responses to protein antigens [45]. The first conserved domain of μ HC (but not δ HC) has been shown to encompass an asparagine-linked glycosylation site at position 46 (N46) that is indispensable for the assembly of a signaling-competent precursor BCR (pre-BCR) in pre-B cells from the mouse bone marrow [48].

The pre-BCR complex includes a μ HC that assembles with the surrogate light chain composed of germline-encoded VpreB and λ 5 polypeptides. Ligand-independent signals from the invariant CD79a (Ig α) and CD79b (Ig β) subunits of the pre-BCR activate pre-B cells through an SLP-65-regulated proliferation and differentiation program, leading to the induction of LC gene recombination and development of immature B cells expressing surface IgM (Fig. 1) [48–54]. In an elegant mouse study, Ubelhart et al. performed

retroviral pre-BCR reconstitution in a triple knockout (TKO) pre-B cell line which lacks SLP-65 (also termed BLNK), RAG2 and λ 5, and is therefore unable to generate endogenous BCRs. They found that pre-BCR μ , but not pre-BCR δ , was capable of inducing ligand-independent calcium mobilization [48]. Remarkably, swapping of N46 from μ HC to δ HC rescued autonomous signaling from pre-BCR δ in reconstituted TKO pre-B cells [48]. In agreement with these results, the authors of this study show a partial developmental block of pre-B cells in μ HC-deficient (i.e., IgM-deficient) mice, which results in the contraction of the early immature B cell compartment [48].

Two earlier studies showing unchanged B cell development in IgM-deficient mice [43, 46] somewhat contradict the findings of Ubelhart et al. [49]. Discrepancies could stem from the use of different mouse models and phenotypic readouts. In the first study [43], IgM-deficient transgenic mice express a restricted HEL-reactive IgD repertoire in the context of a conventional LC-encoding transgene, which has been shown to rescue B cell development in λ 5-deficient mice lacking pre-BCRs [55]. The mechanism behind this rescue remains unclear, but may relate to premature L chain gene recombination. In the second study [46] Lutz et al. (1998) demonstrate that IgM-deficient mice express a diverse IgD repertoire and are capable of fully supporting the development of FO and MZ B cells (B-2). However, the authors found that the levels of B-1 cells expressing elevated surface IgM but little IgD were decreased in these mice. Partial loss of B-1 cells in IgM-deficient mice has also been confirmed by a later study [56]. Taken together these earlier findings raise the possibility that signaling from pre-BCR μ is dispensable for the development of peripheral B-2 cells but not of B-1 cells. Despite the fact that both Lutz et al. (1998) and Ubelhart et al. (2010) used the same mouse strain, the frequency of pre-B cells was reported to be conserved in the initial study [46], and reduced in the later work [48]. Clearly, further work is needed to fully resolve these contradictory results.

Differences between IgD and IgM receptors may further extend to B cells inhabiting secondary lymphoid organs such as the spleen. Mouse studies involving antibody-mediated triggering of surface IgD or IgM have led to the proposal that IgM receptors elicit more splenic B-cell apoptosis but less splenic B-cell proliferation compared with that driven by IgD receptors [57, 58]. Accordingly, studies measuring the intensity and kinetics of BCR-induced protein tyrosine phosphorylation have indicated that signals emanating from surface IgD are stronger and last over a longer timeframe than those emanating from surface IgM [59]. This conclusion was reached from the analysis of mouse J558L myeloma lines transfected with 4-hydroxy-3-iodo-5-nitrophenyl (NIP)-reactive IgD or IgM and subsequently stimulated through NIP-haptenated bovine serum albumin (BSA) [59]. However, no functional differences between IgD and IgM receptors have been observed in a more physiological system involving stimulation of splenic HEL-reactive B cells from BCR transgenic mice by cognate HEL antigen [43, 47].

IgM or IgD receptors on resting B cells are organized as closed oligomeric structures clustered in distinct protein islands, which undergo discrete nanoscale reorganization following

antigenic stimulation [60, 61]. The protein island containing the IgD receptor also includes the BCR co-receptor CD19 and the CXCR4 chemokine receptor responsible for binding CXCL12 [62]. These proteins are functionally connected and indeed CXCR4 signaling as well as CXCL12-mediated B cell migration are impaired in IgD-deficient, but not IgM-deficient mice. These defects could be rescued by signals from CD19 [62]. Overall, this evidence points to the existence of isotype-related specificity in BCR architectures and signaling modalities. The isotype-related specificity of BCR biology can be also concluded from studies of primary mouse B cells showing that IgD has a lower turnover rate than IgM on the B-cell surface, traffics through the Golgi compartment more effectively than IgM, outcompetes IgM for surface display and interaction with CD79a, and forms two distinct complexes with CD79a [63–66]. The isotype-related specificity of BCRs expressed by resting B cells can be further inferred from the observation that surface IgD and IgM follow developmentally regulated expression patterns. In mice, peritoneal mature B-1 cells as well as splenic transitional and mature MZ B cells show elevated surface IgM but reduced surface IgD, whereas conventional mature FO B cells (belonging to the B-2 lineage together with MZ B cells) exhibit increased surface IgD compared to IgM [29, 30].

Another isotype-specific property of BCRs relates to the presence of a glycosyl-phosphatidylinositol (GPI)-linked isoform of the IgD receptor on a fraction of murine B cells [67, 68]. This isoform is not canonically linked to the plasma membrane by invariant Ig α and Ig β proteins [67, 68], which are instead required for the assembly of conventional transmembrane IgD and IgM receptors [69]. Additional mouse studies indicate that GPI-linked IgD receptors are constitutively localized to membrane raft microdomains that drive B cell-activating cyclic AMP signals in response to antigen [67]. Such Ig α / β -independent signals generate B cells with enhanced germinal center seeding capacity [67]. In contrast, canonical Ig α / β -dependent signals from transmembrane IgD receptors generate B cells with attenuated germinal center seeding capacity [67]. By cooperating and interacting with Ig α / β -dependent IgD receptors, GPI-linked IgD receptors may trigger germinal center B cell responses that vary with the strength of antigen. Antigenic conditions that limit receptor occupancy, including reduced antigen concentration or reduced antigen affinity, would selectively stimulate B cells with attenuated germinal center seeding capacity through Ig α / β -dependent signals from transmembrane IgD receptors [67]. In contrast, antigenic conditions determining elevated antigen concentration or elevated antigen affinity would induce B cells with enhanced germinal center seeding capacity through both Ig α / β -independent and Ig α / β -dependent signals from GPI-linked and transmembrane IgD receptors., respectively [67].

IgD receptor and B cell anergy

Elevated surface IgD and reduced surface IgM expression are also hallmarks of anergic B cells present in human peripheral blood and mouse peripheral lymphoid organs, including the spleen [21–25].

Unlike clonal deletion and receptor editing, which eliminate self-reactive B cells in the bone marrow [70–74], clonal anergy allows accumulation of autoreactive B cells in peripheral secondary lymphoid tissues by imposing a state of functional unresponsiveness against self-antigens [26]. Initially described in seminal studies involving transgenic mice expressing a restricted BCR repertoire specific for endogenously expressed HEL [21, 22], anergic B cells with high surface IgD but low surface IgM have been subsequently detected and functionally characterized in reporter mice harboring a diverse BCR repertoire and expressing green fluorescent protein (GFP) under the control of the regulatory region of the Nur77 nuclear orphan receptor [25]. In these Nur77-GFP mice, most transitional and follicular B cells induce GFP expression through BCR-dependent signaling pathways that include SYK kinase and become active in the absence of immunization [25]. Compared to low-GFP-expressing B cells, high-GFP-expressing B cells show decreased surface IgM expression, elevated surface IgD expression, dampened calcium mobilization following antibody-mediated surface IgM ligation, and increased secretion of anti-nuclear autoantibodies in response to polyclonal stimulation [25]. This anergic state has been recapitulated in Nur77-GFP mice crossed with transgenic mice expressing a restricted HEL-reactive BCR repertoire [25]. When exposed to soluble monovalent HEL, peripheral B cells from Nur77-GFP mice up-regulate GFP and acquire anergic features, including low IgM expression combined with elevated IgD expression [25]. Besides identifying Nur77 as a novel hallmark of anergy, these findings indicate that anergy reflects a physiological developmental step resulting from exposure of peripheral B cells to endogenous self-antigens [25]. A possible implication of this discovery is that a certain level of autoreactivity promotes positive selection of B cells by supplying survival and/or developmental signals in peripheral lymphoid organs [56]. Of note, the emergence of physiologically autoreactive B cells with anergic properties from the bone marrow is also common in humans. Indeed, up to 75% of newly generated B cells and 35% of mature B cells present in the human circulation recognize self-antigens, express elevated surface IgD but not IgM, and show some degree of anergy [23, 24, 75].

IgD receptor and tolerance induction

But how do BCRs generate tolerance while retaining responsiveness to foreign antigens? In agreement with earlier studies [21, 22], both IgD expression and IgD signaling function on peripheral B cells from Nur77-GFP mice have been shown to remain intact in anergic B cells [25]. It is possible that IgD outcompetes IgM for association with CD79a and CD79b proteins, which are required for BCR trafficking through the trans-Golgi network, BCR localization on the plasma membrane, and BCR signaling [63, 64]. A recently published study indicates that the retention of stimulating IgD receptors, combined with the down-regulation of anergizing IgM receptors, are critical for the accumulation and persistence of dormant autoreactive B cells in the peripheral lymphocyte repertoire [26]. This conclusion has been

reached through the analysis of two elegant mouse models. In the first model, mice expressing HEL-reactive BCRs have been crossed with *Zfp318*-deficient mice, in which IgD no longer competes with IgM for expression on B cells [26]. In the second model, mutant IgD from *dmit* mice attenuate IgM expression without being assembled on B cells [26]. Analysis of these mice has revealed that IgD ligation by self-antigen actively promotes splenic accumulation of autoreactive B cells independently of IgD-mediated attenuation of surface IgM expression [26]. This process entails IgD-mediated mitigation of an IgM-induced anergy program that promotes surface B-cell up-regulation of the proteoglycan syndecan 1, and intracellular B-cell up-regulation of the transcription factors EGR1 and EGR2 [26]. Of note, surface IgD is competent to autonomously induce the anergy program when expressed in the absence of surface IgM [26], which highlights the dual tolerogenic and B-cell-stimulating functions of IgD receptors.

Besides defining the transcriptional landscape of anergic B cells and identifying syndecan 1 as a novel surface hallmark of anergy [26], this evidence indicates that IgD receptors permit physiologically autoreactive B cells to accumulate in peripheral lymphoid organs under a functional state characterized by unresponsiveness to autologous antigens but responsiveness against foreign antigens (Fig. 4). Indeed, adoptive transfer experiments have shown that anergic IgD⁺IgM^{low} B cells give rise to twice as many germinal center progeny as control naïve IgD⁺IgM^{high} B cells in response to antigen [26, 27]. Consistent with this conclusion, additional evidence obtained from both human and mouse studies indicates that, after entering the germinal center, anergic B cells shift away from self-reactivity while gaining protective reactivity against foreign antigens, including tetanus toxoid, vaccinia virus and rotavirus [26, 27, 76]. This “redemption” program combines N-linked glycosylation of antigen-binding V_H regions with V_HDJ_H gene SHM [26, 27, 76], an AID-mediated process that fosters antigen-driven selection of germinal center B cell mutants expressing antibodies with higher affinity for antigen [77].

One point that remains unclear is how surface IgD stimulates B cells while mitigating IgM-induced anergy. Competition of co-expressed IgD and IgM receptors for specific plasma membrane domains, cytoskeleton-regulated co-receptors and/or intracellular signaling proteins may play an important role [61, 64, 65, 67, 78–80], but intrinsic structural differences between IgD and IgM receptors could also be involved (Fig. 4). In a study supporting this last possibility, a TKO pre-B cell line lacking RAG2, SPL-65 and λ5 was reconstituted with IgM or IgD receptors specific for NIP-haptenated BSA or HEL [56]. In this system, low-valence soluble BSA and HEL instigate calcium-mobilizing signals from IgM but not IgD receptors, whereas polyvalent multimeric BSA and HEL elicit calcium mobilizing signals from both IgM and IgD receptors [56]. Remarkably, the unresponsiveness of IgD receptors to monovalent antigens has been recapitulated in cultures of purified HEL-reactive splenic B cells from BCR transgenic mice and found to be highly dependent on the hinge region (HR) of IgD, which is longer and more flexible than the HR of IgM [56, 81, 82].

While antibodies with short HR have a traditional Y shape, antibodies with a longer HR are more flexible and can acquire

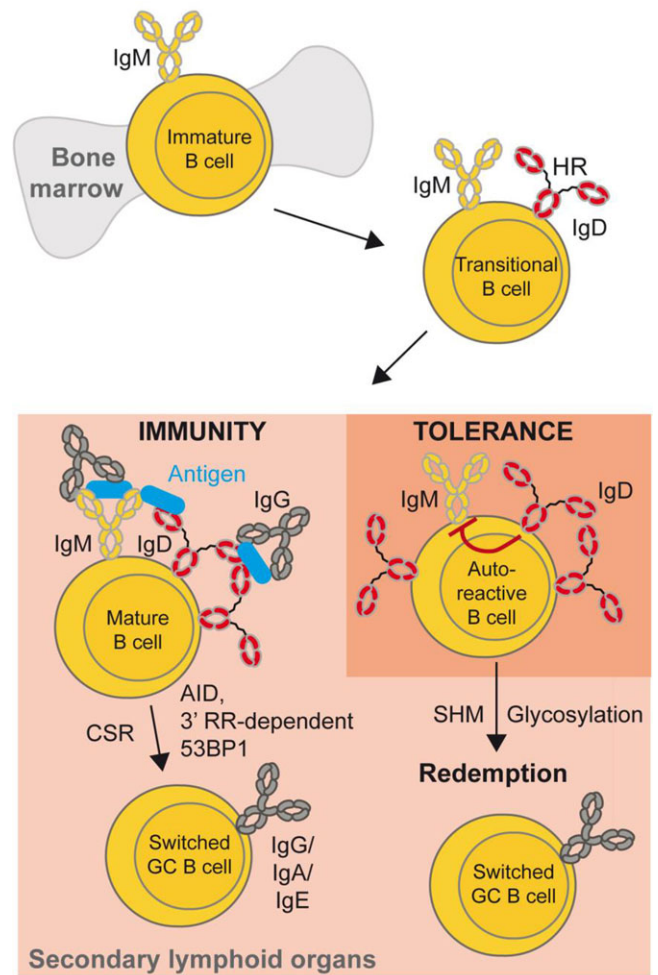


Figure 4. IgD receptors regulate peripheral B cell immunity and tolerance. **IMMUNITY.** Immature B cells expressing surface IgM but not IgD differentiate into transitional B cells. Up-regulation of ZFP318 progressively increases IgD expression on transitional and mature B cells that colonize peripheral lymphoid organs, including the spleen. While IgM receptors have a short hinge region (HR) and predominantly recognize monovalent antigens, IgD receptors have a long HR that may maximize recognition of foreign antigens included in multimeric IgG-containing complexes. Signals from surface IgD promote humoral immunity and memory by stimulating a germinal center program involving classical CSR from IgM to IgG, IgA or IgE, which requires AID, the 3'RR of the IgH locus, and the DNA repair protein 53BP1. **TOLERANCE.** Mature B cells inhabiting peripheral lymphoid organs physiologically retain a certain degree of autoreactivity, but remain unresponsive to IgM engagement by self-antigen through an IgD-regulated state termed anergy. Engagement of surface IgD by self-antigens triggers a signalling program that mitigates IgM-mediated anergy while promoting B cell survival. In the presence of foreign antigens, signals from surface IgD remove autoreactivity from mature B cells by fostering a BCR-editing “redemption” program that entails induction of SHM and BCR glycosylation within germinal centers. Redeemed B cells further undergo CSR and thereafter generate high-affinity IgG (or IgA or IgE) against foreign antigens.

a T shape with two antigen-binding V regions swiveling at the two sides of the C region [81, 83]. This flexibility may potentially contribute to the preferential responsiveness of IgD receptors to multimeric antigens. Remarkably, HR deletion renders surface IgD responsive to monovalent antigens, whereas swapping

the HR of IgD on IgM increases surface IgM responsiveness to polyvalent antigens [56]. Thus, the inherent structural properties of IgD and IgM receptors may regulate B-cell responsiveness to different forms of antigen [84]. In particular, the increased surface IgD/IgM ratio in splenic FO B cells may dampen BCR responses to soluble self-antigens in addition to optimizing BCR responses to foreign multivalent antigens (Fig. 4), including immune complexes [56, 84]. Conversely, the decreased surface IgD/IgM ratio of B-1 and MZ B cells may optimize innate-like antibody responses to monovalent antigens—such as certain phospholipids—shared by both autologous and microbial cells [56, 84].

This simple model is attractive, but needs to be reconciled with recently published results from Sabouri et al. showing prompt responsiveness of surface IgD to monovalent HEL antigen [26]. Sabouri et al. (2016) employed a BCR transgenic system similar but not identical to that used in earlier studies demonstrating the unresponsiveness of surface IgD to monovalent HEL [56]. In the more recent study [26], transgenic mouse B cells expressing HEL-reactive IgD or IgM receptors from mixed splenocyte suspensions were stimulated overnight with a high concentration of soluble HEL (340 μ M). Under these culture conditions, soluble HEL may generate an excess of multivalent complexes for multiple reasons. First, multimeric HEL has been shown to prevail over monomeric HEL at concentrations of soluble HEL close to 350 μ M [85]. Second, soluble HEL is normally present as a reversible dimer within a pH range from 5 to 9, but may undergo multimerization following changes of pH that may occur during overnight cultures [86]. Third, soluble HEL may form multimers by interacting with lysozyme-binding proteins such as α 2-macroglobulin from myeloid and/or stromal cells included in mixed splenocyte cultures [87, 88]. Fourth, some soluble HEL may form multimers by interacting with small amounts of HEL-reactive IgM and IgD released by transgenic B cells spontaneously or in response to activation signals from T cells, myeloid cells and/or stromal cells from mixed splenocyte cultures. In summary, subtle in vitro processes modifying the physical properties of an initially monovalent HEL antigen may lead to the de novo generation of a HEL multimer capable of stimulating surface IgD.

However, it is unlikely that the functional equivalence of transmembrane IgD and IgM simply reflects spontaneous HEL multimerization [26]. Indeed, engagement of transgenic IgD or IgM receptors by nanomolar concentrations of HEL in dose-response in vitro assays induced comparable B cell expression of the activation antigen CD86 [26]. Similarly, engagement of transgenic IgD or IgM receptors by nanomolar concentrations of HEL also induced comparable B cell expression of anergy-related *Egr2* and *Egr3* transcripts in in vivo experiments [26]. Moreover, the reported unresponsiveness of surface IgD to monovalent HEL may be exaggerated in BCR-reconstitution studies involving the TKO pre-B cell line, because this cell system lacks the signaling protein SLP-65 [50, 53, 56]. Further work is clearly needed to resolve these discrepancies and areas of controversy. Nonetheless, it is clear that IgD and IgM receptors establish a complex interplay that regulates self-tolerogenic versus immunogenic fate decisions in peripheral B cells (Fig. 4).

Function of secreted IgD

Circulating human IgD likely derives from class-switched IgM⁺ IgD⁺ plasma cells that enter the circulation after being generated in the lymphoepithelial tissue of aerodigestive organs, including palatine and pharyngeal tonsils [12, 34]. Compared to IgM, IgG and IgA, which have half-lives of about 10, 20 and 6 days, respectively, IgD has a short half-life of about 3 days and is the second least abundant of the human circulating antibodies after IgE [89]. The serum concentration of IgD shows broad variability among individuals, possibly due to a combination of genetic and environmental factors, including exposure to allergens and pollutants [89].

Several lines of evidence support the involvement of secreted IgD in mucosal immunity (Fig. 5). First, compared to plasma cells secreting IgG, IgA or IgM, plasma cells secreting IgD are rare in the human spleen, intestine and bone marrow, but can be readily detected in the human tonsillar mucosa, which may actively stimulate IgM-to-IgD CSR as a result of continuous exposure to commensal bacteria, airborne antigens and food antigens [9, 12]. As noted earlier, some of these tonsillar IgD-secreting plasma cells enter the general circulation to colonize distal mucosal districts, including the middle ear as well as lachrymal, salivary and mammary glands [4, 12, 90]. Second, secreted IgD binds commensals and pathogens inhabiting human upper respiratory and middle ear mucosae, including *Moraxella catarrhalis* and *Haemophilus influenzae* [12, 91]. Third, specific or total IgD increases in the serum of individuals exposed to respiratory pathogens such as rubella virus and *Mycobacterium tuberculosis* or from patients with chronic lung inflammation, including asthma [89]. Fourth, nasal plasma cells secreting IgD increase in IgA-deficient patients with frequent airway infections [92].

Given that IgA is a dominant mucosal antibody required to control both commensals and pathogens [41], IgD responses may increase in IgA-deficient individuals as a compensatory mechanism to preserve mucosal homeostasis. Consistent with this hypothesis, IgD secretion increases in 53BP1-deficient mice lacking IgA (and IgG) production due to impaired classical CSR [14]. As mentioned earlier, IgD from human IgM⁺ IgD⁺ B cells is highly hypermutated [9–11], which may reflect persistent participation of this unique tonsillar B cell subset in sustained germinal center responses to local mucosal antigens (Fig. 4). IgD from human IgM⁺ IgD⁺ B cells is also enriched in autoreactivity [10, 11], which raises the possibility that SHM mitigates IgD reactivity for autologous antigens but enhances IgD reactivity for foreign antigens, including pathogens [12, 14, 26, 76]. Consistent with this idea, secreted IgD from IgM-deficient mice has been shown to enhance protection against vesicular stomatitis virus [46].

Besides binding microbes through both V-specific and C δ -specific mechanisms [4, 12, 14, 91, 93, 94], human secreted IgD establishes an Fc receptor-independent interaction with basophils, mast cells, monocytes and myeloid dendritic cells of the innate immune system [12, 15, 16]. This interaction may involve an evolutionarily conserved but still elusive IgD receptor or receptors (Fig. 5), whose existence is alluded to by the fact that IgD also

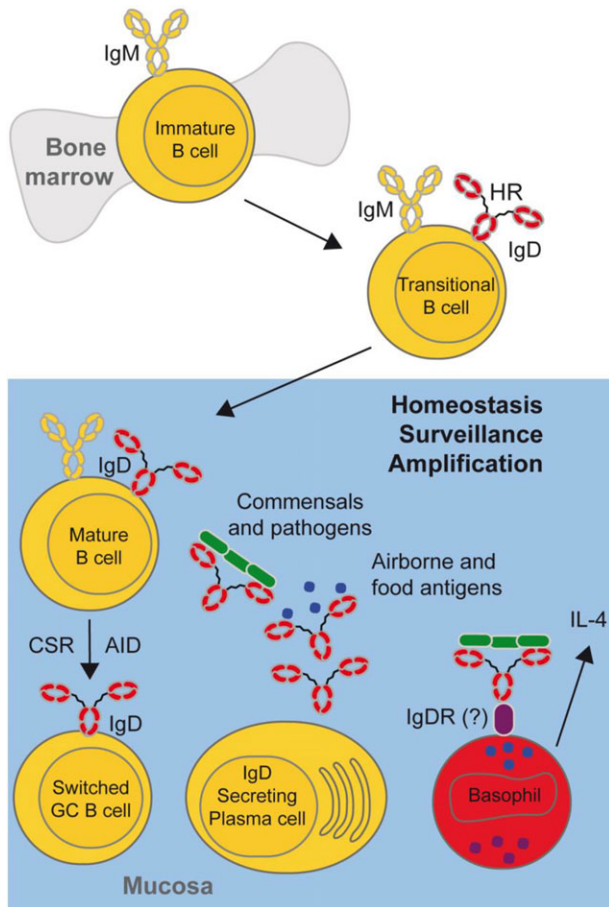


Figure 5. IgD secretion enhances mucosal immune surveillance. Transitional B cells emerging from the bone marrow differentiate into mature FO B cells that colonize the human aerodigestive mucosa, the mouse nasal-associated lymphoid tissue, and mouse mesenteric lymph nodes. At these sites, some FO B cells enter a germinal center program that fosters class switching from IgM to IgD through a non-classical CSR pathway that requires AID but neither the 3'RR nor 53BP1. The resulting IgD class-switched B cells differentiate into plasma cells that release IgD both locally and in the circulation. By recognizing non-invasive commensals, invasive pathogens and probably airborne and food antigens, secreted IgD may enhance immune homeostasis, tolerance and surveillance at mucosal sites of antigen entry. Besides binding antigen in the fluid phase, secreted IgD interacts with mucosal and circulating basophils through an elusive IgD receptor (IgDR). Engagement of basophil-bound IgD by antigen would then trigger basophil release of IL-4, which may amplify humoral immunity by enhancing B cell production of IgG. This antibody is abundantly produced in human nasopharyngeal cavities and may cooperate with IgD to provide local immune protection.

binds a subset of granulocytes from teleosts [12, 36]. In humans, cross-linking of basophil-bound IgD by an isotype-specific antibody triggers basophil release of the B cell-stimulating cytokine IL-4 and thereafter elicits IgG production by B cells [12]. Additional evidence supports the possibility that secreted IgD “arms” basophils with an adaptive recognition system capable of amplifying B cell responses to IgD-targeted antigens via IL-4 (Fig. 5). First, secreted IgD deploys robust immune-enhancing signals in mice through a mechanism seemingly involving the C δ region [95, 96]. Second, treatment of mice with an IgD-cross-linking antibody activates

basophils, induces IL-4 and stimulates massive polyclonal production of IL-4-controlled antibody classes, including IgG1 and IgE [28, 62, 97, 98]. Third, cross-linking of basophil-bound IgG by cognate antigen induces basophil release of IL-4 to amplify antigen-specific IgG production [99]. Clearly, more studies are needed to identify the elusive IgD receptor(s) and to mechanistically dissect how secreted IgD modulates innate and adaptive immune responses.

Concluding remarks

First discovered in 1965 [100], IgD has remained an enigmatic antibody over the past five decades. Recent studies have shown that IgD is evolutionarily conserved from fish to humans and plays a central role in the regulation of tolerogenic and protective B cell responses. Additional findings suggest that IgD belongs to an ancient form of immune surveillance strategically positioned at mucosal sites of antigen entry, including nasopharyngeal cavities. IgD may regulate mucosal homeostasis by establishing a mutualistic interaction with the commensal microbiota at these sites. Future studies should identify and functionally characterize the microbial communities targeted by secreted IgD to clarify how mucosal IgD responses regulate symbiotic host-microbiota interaction. This interaction may foster immune adaptations following post-natal exposure of the aerodigestive mucosa to airborne and food antigens, including allergens. An early dysregulation of these adaptation processes could contribute to the pathogenesis of common disorders such as allergies by disrupting mucosal homeostasis. Finally, it would be important to clarify how and why IgD secretion increases in hyper-IgD syndrome, a rare clinical entity caused by mevalonate kinase deficiency and characterized by periodic fever and spontaneous inflammation [101, 102]. Given that mevalonate kinase is a key enzyme of the cholesterol biosynthetic pathway, lipid mediators emerging from this pathway might regulate IgM-to-IgD CSR and/or IgD secretion.

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Abbreviations: AID: activation-induced cytidine deaminase · BCR: B cell antigen receptor · C: constant region gene · CSR: class switch recombination · D: diversity gene · FO: follicular · GC: germinal center · H: heavy chain · HR: hinge region · Ig: immunoglobulin · J: joining gene · L: light chain · MZ: marginal zone · SHM: somatic hypermutation · V: variable gene · ZFP318: zinc-finger protein 318

Full correspondence: Dr. Andrea Cerutti, Institut Hospital del Mar d'Investigacions Mèdiques (IMIM) at Barcelona Biomedical Research Park (PRBB), Avenida Dr. Aiguader 88, Barcelona 08003, Spain
Fax: +34-933160410
e-mail: acerutti@imim.es

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